Variational Autoencoders
Experimental Report

Detailed documentation of implementation steps, experimental results, and conclusions for VAE-based generative models.

May 25, 2026

Report Structure

- **Part I — Background:** Theory of VAEs and model architecture
- **Part II — Experimental Setup:** Fixed hyperparameters and dataset
- **Part III — Experiments:** Per-experiment results, analysis, and figures (losses, samples, reconstructions, latent spaces)
- **Part IV — Conclusions:** Summary of findings and key differences

Contents

1 Introduction

This report documents a systematic experimental study of **Variational Autoencoders (VAEs)** applied to image generation. Each experiment targets a specific design choice — conditioning strategy, attention mechanism, perceptual loss, or capacity — and evaluates its effect on generation quality, reconstruction fidelity, and latent-space structure.

The document is designed to serve as one half of a broader **comparative analysis between VAEs and Denoising Diffusion Probabilistic Models (DDPMs)**, enabling an apples-to-apples comparison of two dominant paradigms in deep generative modelling.

1.1 Purpose of the Comparison

VAEs and DDPMs represent fundamentally different inductive biases:

- **VAEs** learn an explicit, amortised posterior $q_\phi(\mathbf{z}|\mathbf{x})$ and optimise a tractable Evidence Lower Bound (ELBO), trading reconstruction sharpness for a structured, continuous latent space.
- **DDPMs** learn to reverse a fixed Markov diffusion process, achieving high sample quality at the cost of slow, iterative inference.

Both model families are trained on the **similar architectural backbone with differences we will elaborate later**, the same dataset, so that differences in metrics reflect the generative paradigm rather than implementation disparity.

Note: I tried as much as possible to keep the experiments consistent, but I could not fully achieve that because it requires much more time and additional training.

2 Background: Variational Autoencoders

2.1 Variational Autoencoder (VAE)

A Variational Autoencoder (VAE) is a probabilistic generative model that learns a compact latent representation of data while enabling sampling and generation of new examples. Unlike a standard autoencoder that maps an input directly to a deterministic latent vector, a VAE learns a distribution over the latent space, typically parameterized by a Gaussian distribution with mean μ and variance σ^2.

The encoder maps an input x into latent parameters:

$$q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$$

A latent vector z is then sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

The decoder reconstructs the input from the latent variable:

$$p_\theta(x|z)$$

The objective of the VAE is to jointly optimize reconstruction quality and latent-space regularization.

2.2 KL-Divergence Regularization and Its Role

The VAE loss consists of two main components:

$$\mathcal{L}_{VAE} = \mathcal{L}_{recon} + \beta D_{KL}\left(q_\phi(z|x) \| p(z)\right)$$

The KL-divergence term regularizes the learned latent distribution so that it remains close to a predefined prior distribution, usually:

$$p(z) = \mathcal{N}(0, I)$$

This regularization has several important roles:

- It enforces continuity and smoothness in the latent space.
- It prevents the encoder from memorizing training samples.
- It enables meaningful interpolation and random sampling.
- It improves the generative capability of the model.

Without KL regularization, the latent space may become unstructured, making generation unstable or inconsistent.

2.3 Perceptual / Feature-Matching Losses

Pixel-wise reconstruction losses such as Mean Squared Error (MSE) or L1 loss often produce blurry outputs because they optimize only low-level similarity. To improve perceptual quality, perceptual or feature-matching losses are commonly used.

Instead of comparing images directly in pixel space, the comparison is performed in the feature space of a pretrained neural network:

$$\mathcal{L}_{perc} = \sum_i \|\phi_i(x) - \phi_i(\hat{x})\|_2^2$$

where:

- x is the target image,
- $\hat{x}$ is the reconstructed image,
- $\phi_i(\cdot)$ denotes features extracted from intermediate layers of a pretrained network.

Feature-matching losses encourage the generated outputs to preserve semantic structure, texture, and perceptual realism rather than only exact pixel correspondence.

2.4 Conditional Variational Autoencoder (CVAE)

A Conditional Variational Autoencoder (CVAE) extends the VAE by conditioning both the encoder and decoder on additional information c, such as class labels, text embeddings, or other modalities.

The conditional encoder becomes:

$$q_\phi(z|x, c)$$

and the decoder becomes:

$$p_\theta(x|z, c)$$

Conditioning allows the model to generate outputs that are guided by external information while still benefiting from the generative properties of the latent space.

2.5 Conditional Generation Strategies

2.5.1 Adaptive Group Normalization (AdaGN)

One conditioning strategy explored in this work is Adaptive Group Normalization (AdaGN). I became familiar with this idea while studying CME 296, specifically during Lecture 5, where it was discussed that Adaptive Layer Normalization can outperform Cross-Attention and In-Context Conditioning in the context of Diffusion Transformer (DiT) architectures.

This motivated me to investigate whether similar conditioning principles could be applied to VAEs. The idea is to inject conditioning information into normalization layers by modulating feature statistics using learned scale and shift parameters:

$$\mathrm{AdaGN}(h, c) = \gamma(c)\frac{h - \mu(h)}{\sigma(h)} + \beta(c)$$

where:

- h represents intermediate features,
- $\gamma(c)$ and $\beta(c)$ are conditioning-dependent modulation parameters.

This conditioning mechanism enables stronger control over feature representations while maintaining architectural efficiency.

2.5.2 Concatenation-Based Conditioning

Another conditioning approach explored was simple concatenation. This method was inspired by implementations and discussions found in external resources.

In concatenation-based conditioning, the condition vector c is directly appended to either the input, latent representation, or intermediate feature maps:

$$z' = [z; c]$$

The concatenated representation is then passed through the decoder or subsequent layers.

Although simpler than adaptive normalization methods, concatenation remains effective and easy to implement, making it a useful baseline for conditional generation experiments.

2.6 Model Architecture

3 Experimental Setup

3.1 Dataset

All experiments use **CIFAR-10** (Canadian Institute For Advanced Research) which is a foundational dataset in computer vision used to train and evaluate image generation models. It consists of 60,000 32×32 pixel color images across 10 distinct, mutually exclusive object and animal classes.

3.2 Fixed Hyperparameters

The following hyperparameters are held constant across *all* experiments to ensure fair comparison:

Fixed Hyperparameters (all experiments)

```
{
    "image_size"    : 32,         # Input / output resolution (
       px)
    "num_classes"   : 10,         # Number of conditioning
       classes
    "latent_dim"    : 128,        # Dimensionality of latent
       space z
    "base_channels" : 64,         # Feature-map width at first
       encoder level
    "channel_mults" : [1, 2, 4],# Channel multipliers per
       resolution level
    "num_res_blocks": 2,          # Residual blocks per
       resolution level
    "batch_size"    : 32,         # Training batch size
    "lr"            : 0.0002,     # Adam learning rate
}
```

3.3 Model Variants

- **VAE** — Vanilla VAE with reconstruction + KL loss only.
- **PercepVAE** — VAE augmented with a perceptual (feature-matching) loss.
- **cVAE-AdaGN** — Conditional VAE using Adaptive Group Normalisation to inject class information.
- **cVAE-Concat** — Conditional VAE using channel concatenation of the class embedding.

3.4 Losses

3.4.1 Perceptual loss -Added-

To improve perceptual similarity between the predicted image $\hat{x}$ and the target image x, we employ a *perceptual loss* based on deep feature comparisons extracted from a pretrained convolutional neural network. Unlike pixel-wise losses (e.g., ℓ_1 or ℓ_2), this loss measures differences in a learned feature space that better correlates with human perception.

We use a pretrained VGG-16 network as the feature extractor. Let $\phi(\cdot)$ denote the VGG-16 feature map, truncated at selected intermediate layers. In our implementation, we extract features from multiple layers corresponding to higher-level semantic representations, specifically:

$$\mathcal{L}_{\text{perc}}(\hat{x}, x) = \sum_{l \in \mathcal{S}} \left\| \phi_l(\hat{x}) - \phi_l(x) \right\|_1, \tag{1}$$

where $\mathcal{S} = \{8, 15, 22\}$ corresponds to VGG-16 layers (ReLU activations after blocks 2, 3, and 4), and $\phi_l(\cdot)$ denotes the feature representation obtained after layer l.

Feature Extraction. The VGG-16 backbone is divided into sequential sub-networks at predefined "tap points." Each sub-network processes the input progressively, allowing intermediate feature maps to be extracted at multiple depths. This enables the loss to capture both low-level texture information and high-level semantic structure.

Input Normalization. Since VGG-16 expects ImageNet-normalized inputs, the input images (originally in $[-1, 1]$) are first rescaled to $[0, 1]$, then normalized using the standard ImageNet statistics:

$$x' = \frac{x + 1}{2}, \tag{2}$$

$$\hat{x}' = \frac{\hat{x} + 1}{2}, \tag{3}$$

$$x_{\text{norm}} = \frac{x' - \mu}{\sigma}, \tag{4}$$

where $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$.

Optimization Details. All parameters of the feature extractor are frozen during training:

$$\nabla_{\theta_\phi} \mathcal{L}_{\text{perc}} = 0, \tag{5}$$

ensuring that only the generator (or predictor network) is updated.

Final Loss. The final perceptual loss is computed as a sum of layer-wise ℓ_1 distances:

$$\mathcal{L}_{\text{perc}} = \sum_{l \in \mathcal{S}} \left\| \phi_l(\hat{x}) - \phi_l(x) \right\|_1, \tag{6}$$

which encourages the generated image to match the target both in low-level details and high-level perceptual structure.

3.5 Evaluation Metrics

FID Fréchet Inception Distance — measures the distributional gap between real and generated images in Inception feature space. *Lower is better.*

IS (Inception Score) Evaluates both the quality and diversity of generated images. Reported as mean ± standard deviation over multiple splits. *Higher is better.*

4 Evaluation Strategy

We have 2 evaluation metrics categories:

- **Reference-free metrics (FID, IS)**
- **Reference-based metrics (MSE, PSNR, SSIM, LPIPS)**

Both **Inception Score (IS)** and **Fréchet Inception Distance (FID)** evaluate generative models by passing generated images through the pre-trained **Inception-v3** network. The fundamental shift between the two is that IS only analyzes the generated images, while FID compares the generated images directly against the real dataset. Because of this architectural difference, FID has almost entirely replaced IS as the modern standard for evaluating GANs and Diffusion models.

4.1 Inception Score (IS)

IS attempts to capture human judgment by measuring two properties simultaneously:

- **Image Quality**
- **Image Diversity**

It does this by analyzing the final class probabilities output by the Inception network.

Quality (Identifiability) A high-quality image should look clearly like a specific object. Mathematically, the conditional probability $P(y|x)$ (the model's prediction for a specific generated image x) should have low entropy. It should be highly confident (e.g., 99% sure the image is a dog).

Diversity The generator should produce a wide variety of objects, not just dogs. The marginal probability $P(y)$ (the average prediction across all generated images) should have high entropy, meaning all ImageNet classes are equally represented overall.

IS mathematically combines these concepts by calculating the KL divergence between the conditional and marginal distributions:

$$\text{IS} = \exp\left(\mathbb{E}_x\left[D_{\text{KL}}\Big(P(y|x) \,\|\, P(y)\Big)\right]\right)$$

The Fatal Flaw of IS IS never references the real training data. If a model memorizes exactly one perfect image from every single ImageNet class and generates nothing else, it will achieve a flawless IS. It completely fails to detect severe mode collapse if the collapsed modes are recognizable objects.

4.2 Fréchet Inception Distance (FID)

Instead of looking at the final class probabilities, FID removes the final classification layer of the Inception network and analyzes the deep continuous features (latent embeddings) from the intermediate layer.

FID assumes these continuous feature vectors follow a multivariate Gaussian distribution. It calculates the mean (μ) and covariance (Σ) for the embeddings of the real dataset (r) and the generated dataset (g), and then computes the Fréchet distance (also known as the Wasserstein-2 distance) between these two distributions:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)$$

4.3 Why FID Wins

- **It uses ground truth:** By comparing fake embeddings directly to real embeddings, it penalizes models that generate high-quality images that do not actually belong to the target distribution.
- **It catches mode collapse:** If a model only generates a limited variety of images, the variance/covariance (Σ_g) of its embeddings will be artificially small. This mismatch with the real variance (Σ_r) severely penalizes the Fréchet distance.
- **Lower is better:** Because it is a true distance metric, a perfect score is 0.0, meaning the generated distribution perfectly overlaps the real distribution.

4.4 IS and FID for Unconditional Generative Models

Neither **Inception Score (IS)** nor **Fréchet Inception Distance (FID)** depends on whether a generative model is conditional or unconditional. Both metrics evaluate only the generated output images, treating the generator itself as a complete black box.

The metrics do not inspect the model architecture, training objective, or conditioning mechanism. They operate solely on the generated samples. This means they work identically for models that generate images purely from random noise.

4.4.1 How Inception Score (IS) Works Unconditionally

A common source of confusion comes from the notation used in IS, specifically the KL divergence between the conditional distribution $P(y|x)$ and the marginal distribution $P(y)$.

In this context, the word "conditional" does not refer to the generative model itself. Instead, it refers to the predictions produced by the pre-trained Inception-v3 classifier.

Extracting $P(y|x)$ An image generated unconditionally from random noise is passed into the pre-trained Inception-v3 network. The classifier outputs a probability distribution over its 1,000 ImageNet classes.

For example, the classifier may predict:

$$P(y|x) = \begin{cases} 0.99 & \text{dog} \\ 0.01 & \text{other classes} \end{cases}$$

This distribution represents the classifier's confidence about the generated image x.

Extracting $P(y)$ The process is repeated for thousands of generated images. The predicted class distributions are averaged to obtain the marginal distribution:

$$P(y) = \mathbb{E}_x[P(y|x)]$$

This reflects the overall diversity of classes produced by the generator.

Interpretation If the unconditional model generates high-quality and recognizable objects, then $P(y|x)$ will be sharp (low entropy). If it generates a diverse variety of objects, then $P(y)$ will be broad and approximately uniform (high entropy). Together, these properties lead to a high Inception Score.

4.4.2 How FID Works Unconditionally

FID is more straightforward because it evaluates continuous feature embeddings rather than class probabilities.

1. Generate a large batch of images unconditionally from random noise.
2. Pass the generated images through the Inception-v3 network and extract feature embeddings from the intermediate layer.
3. Compute the mean vector (μ_g) and covariance matrix (Σ_g) of the generated embeddings.
4. Compare these statistics against the mean vector (μ_r) and covariance matrix (Σ_r) computed from the real training dataset.

The Fréchet Inception Distance is then computed as:

$$\mathrm{FID} = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)$$

A lower FID indicates that the generated feature distribution more closely matches the real data distribution.

5 Experiment 1

5.1 Motivation

The first experiment establishes the **baseline** performance of four VAE variants that differ in their conditioning and loss strategy, without self-attention. This provides a reference point for subsequent ablations.

5.2 Hyperparameters

Experiment 1 — Variable Hyperparameters

```
{
    "epochs"         : 50,
    "kl_weight"      : 0.001,    # Weight on KL-divergence term
    "percep_weight"  : 0.1,      # Weight on perceptual loss
    "use_attention"  : false,    # Self-attention disabled
}
```

5.3 Evaluation Results

Table 1: Experiment 1 — Quantitative evaluation (FID and IS). FID: lower is better; IS: higher is better.

Model	FID ↓	IS mean ↑	IS std
VAE	195.33	2.98	0.09
PercepVAE	174.83	2.93	0.08
cVAE-AdaGN	171.04	3.16	0.10
cVAE-Concat	163.23	3.54	0.10

5.4 Training Dynamics

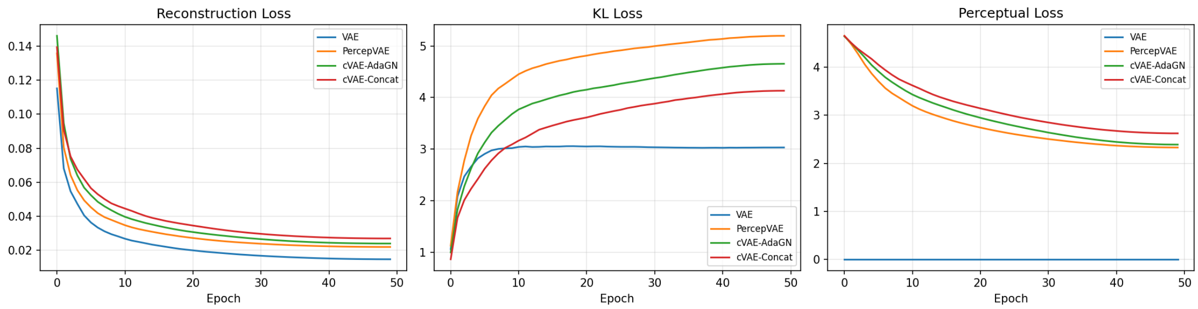


Figure 1: Experiment 1 — Training loss curves for all four VAE variants. The plot shows the evolution of the total loss, reconstruction term, and KL-divergence term over 50 epochs.

5.5 Qualitative Results

5.5.1 Random Samples

Figure 2: Experiment 1 — Randomly sampled images generated by decoding latent vectors $\mathbf{z} \sim \mathcal{N}(0, I)$.

5.5.2 Reconstructions

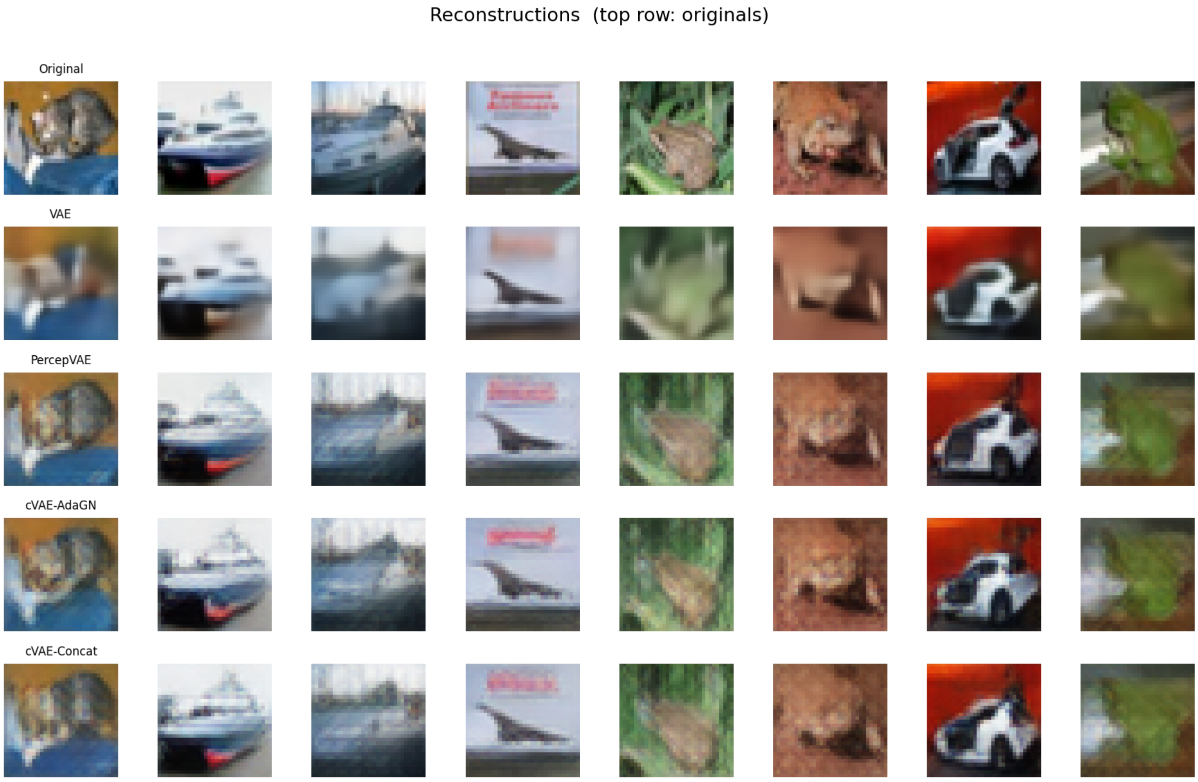
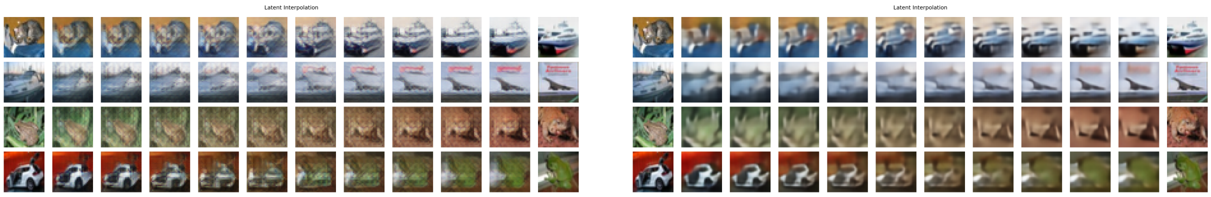


Figure 3: Experiment 1 — Input images (top row) and their reconstructions (bottom row). Reconstruction quality is noticeably better than free generation, reflecting the encoder's ability to infer informative posterior means.

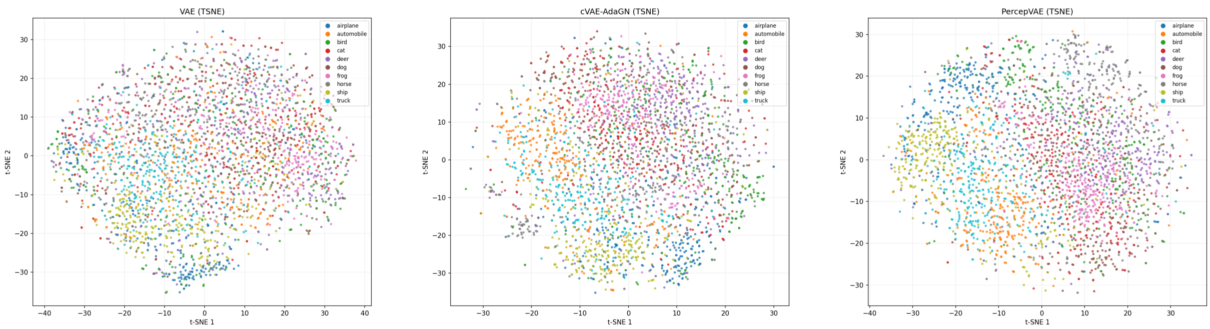
5.5.3 Latent-Space Interpolations



(a) Interpolation For CVAE-AdaGN. (b) Interpolation For Vanilla VAE.

Figure 4: Experiment 1 — linear interpolation between pairs of encoded images in latent space.

5.5.4 Latent-Space Visualisation



(a) VAE (b) cVAE-AdaGN (c) Perceptual VAE

Figure 5: Experiment 1 — 2-D t-SNE projections of the latent space, coloured by class label. Tighter, better-separated clusters indicate that the model has learned class-discriminative structure in $\mathbf{z}$.

5.6 Analysis and Observations

- **Conditioning is decisive.** Both cVAE variants outperform the unconditional baselines by a substantial margin, with cVAE-Concat achieving the lowest FID (163.23) and the highest IS (3.54).
- **Perceptual loss helps FID but not IS**, suggesting it encourages more realistic textures at the possible cost of diversity.
- **The KL loss of the vanilla VAE.** The KL loss of the Vanilla VAE saturates early as we see in Figure 1 and this might explain the differences between the generated noise between VAE and the other model that seems like try to cluster the noise in something meaningful (see Figure 16).
- **Perceptual loss provides moderate benefit.** PercepVAE reduces FID by $\approx$20 points vs. the vanilla VAE, but its IS slightly decreases, suggesting that perceptual regularisation may introduce a diversity-quality trade-off.
- **AdaGN vs. Concat.** cVAE-Concat outperforms cVAE-AdaGN on both FID and IS, indicating that direct feature concatenation is a more effective conditioning mechanism than normalisation-layer modulation at this scale.

- **Reconstruction vs. generation gap.** Reconstruction quality is visually good (see Figure 3), while free-sampling quality remains limited (Figure 16). This posterior-prior gap is a well-known VAE limitation.

6 Experiment 2

6.1 Motivation

As we can see in the first experiment, the generative capability of the model is limited, as it behaves more like an Autoencoder (AE). A probable reason is that the KL divergence regularization term, β, does not effectively contribute to shaping the latent space. Therefore, we set $\beta = 1.0$.

6.2 Hyperparameters

Experiment 2— Variable Hyperparameters

```
{
    "epochs"          : 50,
    "kl_weight"       : 1.0,    # Weight on KL-
       divergence term
    "percep_weight"   : 0.1,     # Weight on perceptual
        loss
    "use_attention"   : false,   # Self-attention
       disabled
    "attn_heads"      : 1,
}
```

6.3 Evaluation Results

Table 2: Experiment 2— Quantitative evaluation (FID and IS). FID: lower is better; IS: higher is better.

Model	**FID** $\downarrow$	**IS mean** $\uparrow$	**IS std**
VAE	326.79	1.69	0.03
PercepVAE	276.51	2.22	0.06
cVAE-AdaGN	394.40	2.03	0.03
cVAE-Concat	193.62	3.41	0.12

6.4 Training Dynamics

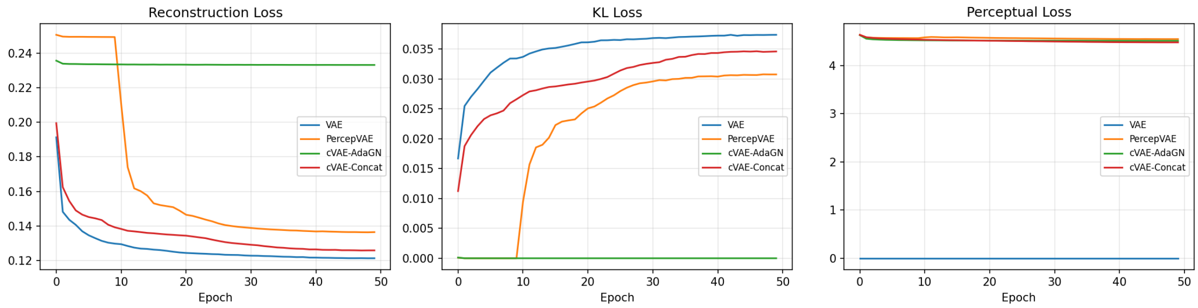


Figure 6: Experiment 2— Training loss curves for all four VAE variants. The plot shows the evolution of the total loss, reconstruction term, and KL-divergence term over 50 epochs.

6.5 Qualitative Results

6.5.1 Random Samples

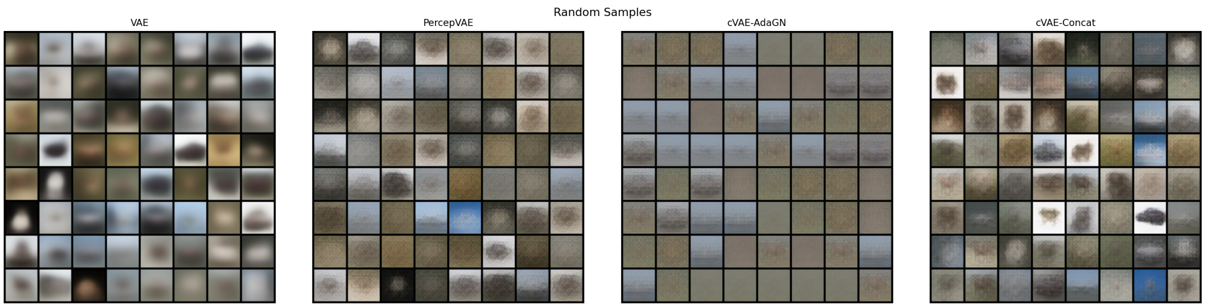
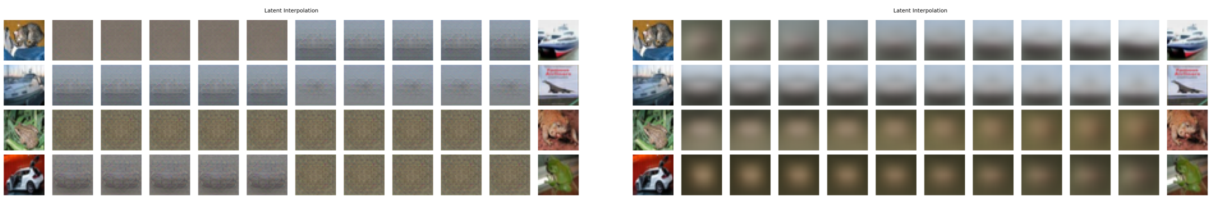


Figure 7: Experiment 2— Randomly sampled images generated by decoding latent vectors $\mathbf{z} \sim \mathcal{N}(0, I)$.

6.5.2 Reconstructions

Figure 8: Experiment 2— Input images (top row) and their reconstructions (bottom row).

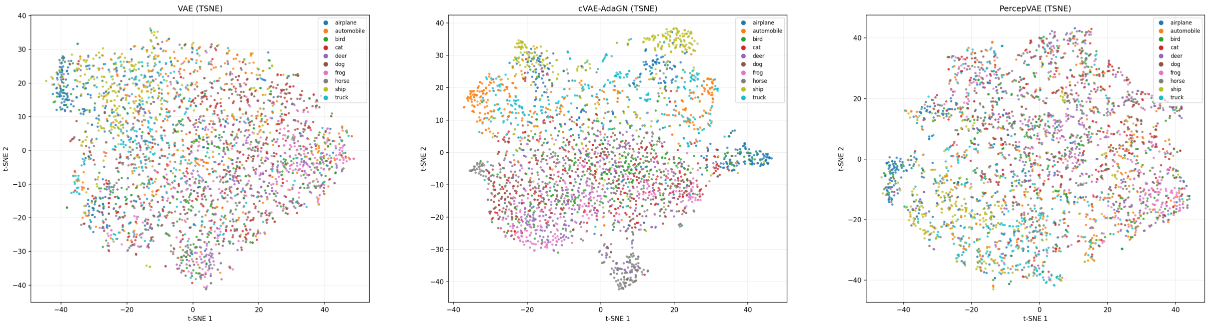
6.5.3 Latent-Space Interpolations



(a) Interpolation For CVAE-AdaGN. (b) Interpolation For Vanilla VAE.

Figure 9: Experiment 2— linear interpolation between pairs of encoded images in latent space.

6.5.4 Latent-Space Visualisation



(a) VAE (b) cVAE-AdaGN (c) Perceptual VAE

Figure 10: Experiment 2— 2-D t-SNE projections of the latent space, coloured by class label. Tighter, better-separated clusters indicate that the model has learned class-discriminative structure in $\mathbf{z}$.

6.6 Analysis and Observations

- **It seems that β is very high.** One of the failure modes in VAE is the posterior collapsing and these result point out that is our case and we can see it clearly in the cVAE-AdaGN, it has the largest FID and in (Figure 6) you can se it has 0.0 KL loss.
- **VAE sill produce blurry images.** PercepVAE reduces FID by ≈20 points vs. the vanilla VAE, but its IS slightly decreases, suggesting that perceptual regularisation may introduce a diversity-quality trade-off.
- **AdaGN vs. Concat.** cVAE-Concat is better than cVAE-AdaGN on both FID and IS, indicating that direct feature concatenation still more effective conditioning mechanism. AdaGN gives the model learning capacity more than the Concatenation as the AdaGN inject weight at each ResBlock but in concatenation we concate only on the latent embedding and I think this is the reason for falling on the posterior collapse.
- **The model performs poorly in both generation and reconstruction.** Unlike the previous experiment, where the reconstruction quality was better than the generation quality, both aspects are poor in this case. This can be observed from the loss curves in Figure 6, especially for the VAE, which shows an almost constant reconstruction loss. This behavior is also evident in the visualizations.
- **Perceptual Loss is saturated and equal for all models** This make sense as all images are blurry so there is no perceptual feature to learn as the KL term is the dominant.

7 Experiment 3

7.1 Motivation

After observing the two extreme values of β, we now choose a moderate value to study its effect. I also applied the same experiment with attention and you can see it in the repo.

7.2 Hyperparameters

Experiment 3— Variable Hyperparameters

```
{
    "epochs"        : 30,    # To accelerate the
        expermenting
    "kl_weight"     : .1,    # Weight on KL-divergence
        term
    "percep_weight" : 0.1,      # Weight on perceptual
        loss
    "use_attention" : False,

}
```

7.3 Evaluation Results

Table 3: Experiment 3— Quantitative evaluation (FID and IS). FID: lower is better; IS: higher is better.

Model	FID ↓	IS mean ↑	IS std
VAE	209.44	2.16	0.04
PercepVAE	121.52	4.77	0.14
cVAE-AdaGN	118.01	4.85	0.19
cVAE-Concat	112.86	5.07	0.19

7.4 Training Dynamics

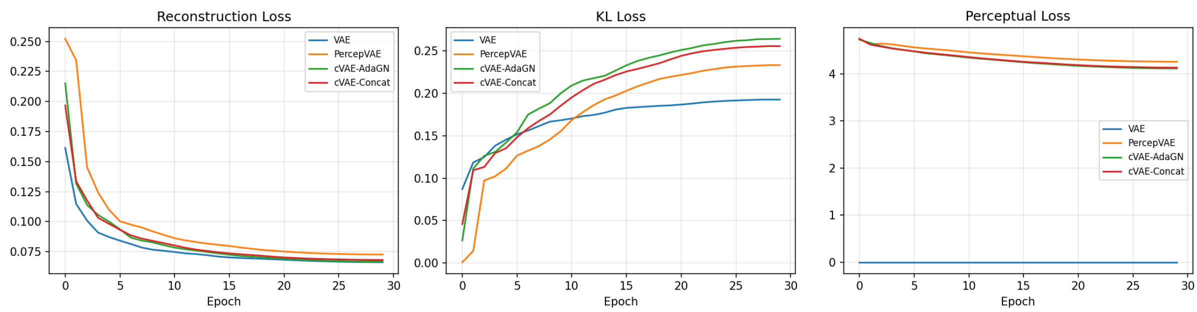


Figure 11: Experiment 3— Training loss curves for all four VAE variants. The plot shows the evolution of the total loss, reconstruction term, and KL-divergence term over 50 epochs.

7.5 Qualitative Results

7.5.1 Random Samples

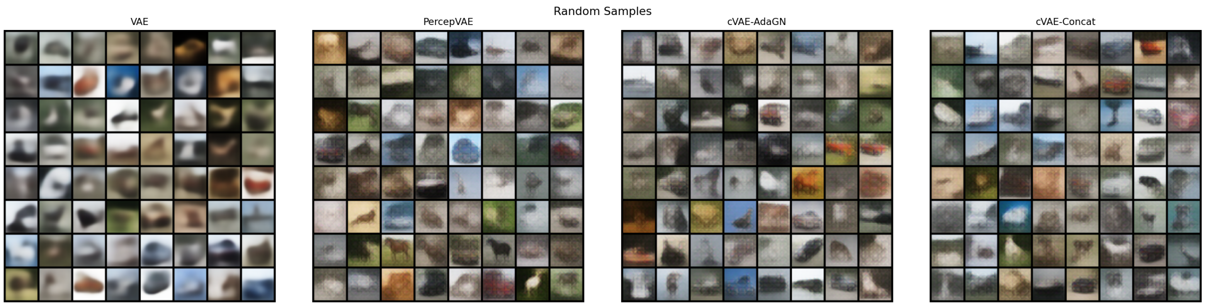


Figure 12: Experiment 3— Randomly sampled images generated by decoding latent vectors $\mathbf{z} \sim \mathcal{N}(0, I)$.

7.5.2 Reconstructions

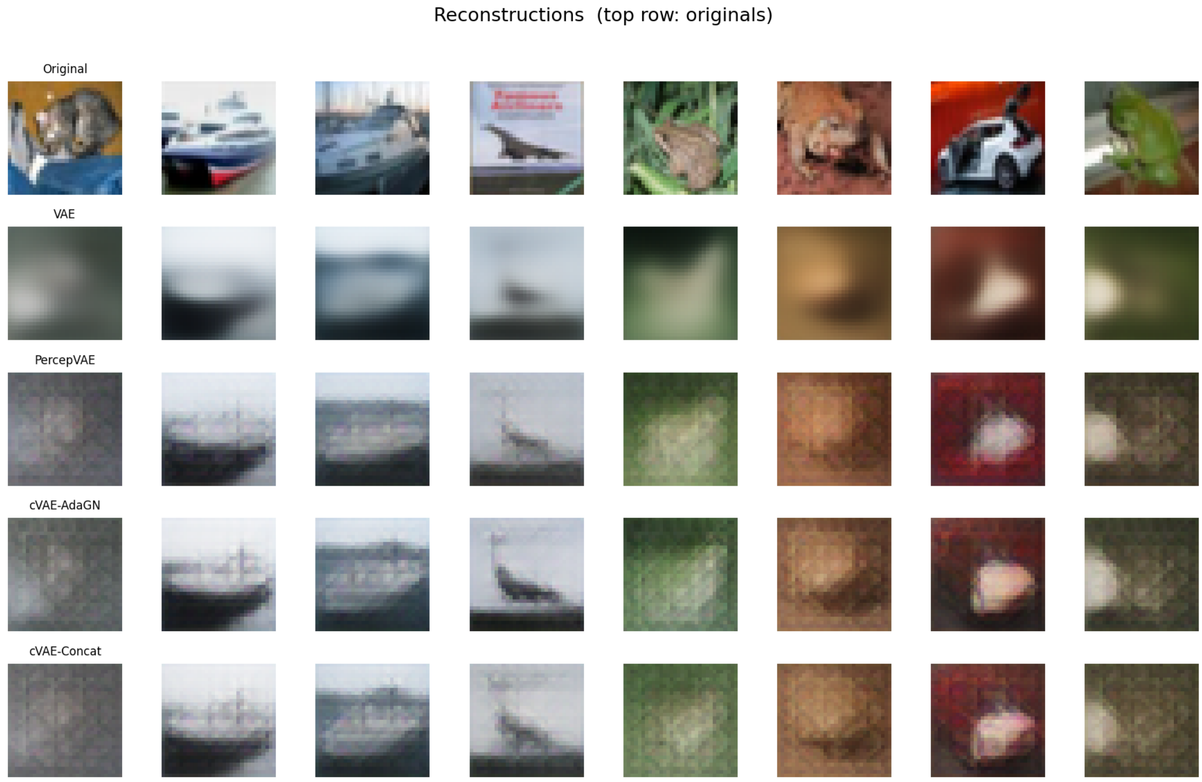
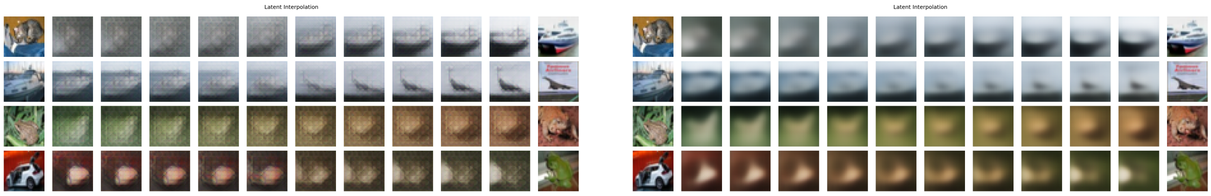


Figure 13: Experiment 3— Input images (top row) and their reconstructions (bottom row).

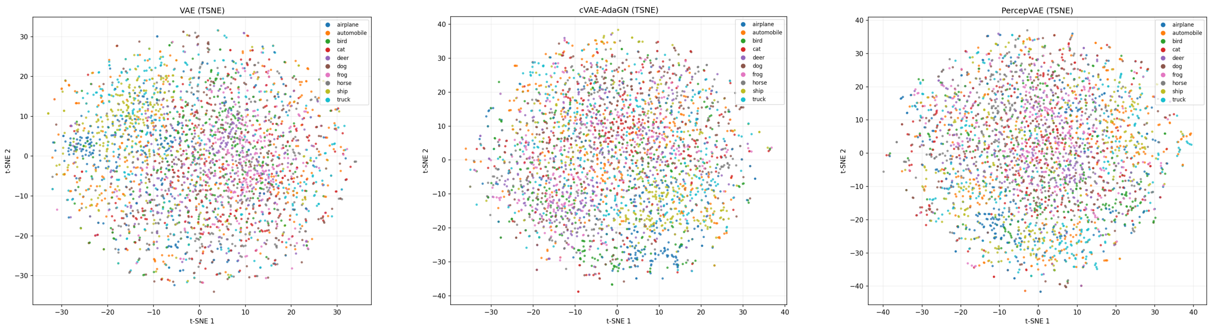
7.5.3 Latent-Space Interpolations



(a) Interpolation For CVAE-AdaGN. (b) Interpolation For Vanilla VAE.

Figure 14: Experiment 3— linear interpolation between pairs of encoded images in latent space.

7.5.4 Latent-Space Visualisation



(a) VAE (b) cVAE-AdaGN (c) Perceptual VAE

Figure 15: Experiment 3— 2-D t-SNE projections of the latent space, coloured by class label. Tighter, better-separated clusters indicate that the model has learned class-discriminative structure in $\mathbf{z}$.

7.6 Analysis and Observations

- **Whoa!** Finally, we are able to see something meaningful—though it is still blurry—but we have succeeded.
- **VAE sill produce blurry images.** PercepVAE reduces FID by ≈20 points vs. the vanilla VAE, but its IS slightly decreases, suggesting that perceptual regularisation may introduce a diversity-quality trade-off.
- **AdaGN vs. Concat.** cVAE-Concat is better than cVAE-AdaGN on both FID and IS, indicating that direct feature concatenation still more effective conditioning mechanism.
- **The model performs moderately in both generation and reconstruction.** Here we can see that the generative ability get better and it's our goal.
- **Perceptual Loss is saturated and equal for all models** This make sense as all images are blurry so there is no perceptual feature to learn as the KL term is the dominant partially, we will discuss later.

- **The KL loss of vanilla VAE** Here er can see other wise the previous experiments that the KL loss of the VAE is not saturated/constant, we will discuss the behavior of the KL loss later.
- **The latent Interpolation** We can see that the transition in smoother between the classes.

8 Cross-Experiment Discussion

8.1 Summary Table — Best Model

Table 4: Summary of the best-performing VAE variant per experiment.

Exp.	Best Model	Key Change	FID ↓	IS ↑
1	cVAE-Concat	Moderate β and attention	112.86	5.7

8.2 Discussion

- **Blurriness everywhere** MSE loss measures the average squared difference between predicted and true values. It trains models to minimize pixel-wise errors.

  When outputs are uncertain (like images), MSE makes the model learn the average of all possible correct outputs.

  We can observe this clearly when using horizontal flipping as augmentation. Ideally, the flipped image should have the same embedding as the original image; So, when decoding it, we sometimes obtain distorted outputs, such as a horse with two heads—something resembling a “CatDog”/"BesbesBoby" effect but here is HorseHorse.

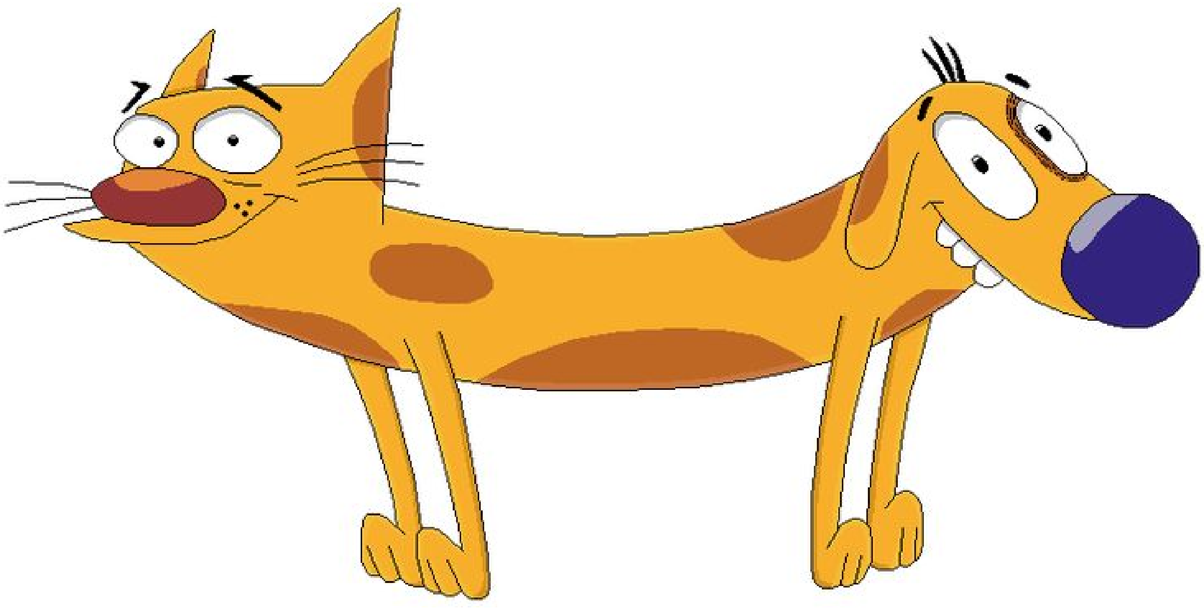


Figure 16: CatDog

This averaging removes high-frequency details (edges, textures), so the result looks blurry instead of sharp.

- **Perceptual Loss saturated** I think this is due to the blurriness of the images; when we feed them into the model, they produce very similar feature maps each time, leading to the same error. However, I will investigate this further later. But we can see its effect in Experiment1 when we reconstruct the images but for the later two Experiments it has minor effect. I think the competition between the loss terms is also a reason for this saturation.

- **There is like Checkboard Artifact** At first, I used transposed convolution (Conv2DTranspose) for upsampling and it suffer from this artifact, but I later changed it to nearest-neighbor interpolation followed by a convolutional layer. However, the same artifacts still appeared. I also tried bilinear interpolation, but the artifacts were still present.

  While studying CME 296 at Stanford, I learned that the weighting of the perceptual loss can sometimes cause these kinds of artifacts. Due to time constraints, I was unable to include perceptual loss in my experiments.

- **KL Term always Diverges** At the start of training, the encoder is "lazy" — it outputs distributions very close to $\mathcal{N}(0, 1)$ (low KL), but this means it encodes almost no useful information about the input. The reconstruction loss is therefore very poor.

  As training progresses, the encoder begins to encode real structure about the data into the latent space: the means shift away from 0, and the variances move away from 1. As a result, the KL divergence increases, but the reconstruction quality improves.

  This creates a tug-of-war:

  - Reconstruction loss pushes the model to encode more information (increasing KL).
  - KL loss pushes the model to stay close to $\mathcal{N}(0, 1)$ (decreasing KL).

- **why does the conditional model outperform the unconditional one?**

  The unconditional model tries to capture the full uncertainty of the data distribution, which is a difficult task for datasets like CIFAR-10, where data points vary extensively and the overall uncertainty is high.

  In contrast, conditioning allows the model to reduce this uncertainty by restricting it within each class, where samples share more common features. In our case, this is still challenging because objects within the same class can appear in many poses and conditions. However, for simpler datasets like MNIST, conditioning is typically more effective and leads to better performance.

- **Why does concatenation-based conditioning perform better than AdaGN-based conditioning?**

  Concatenation tends to introduce conditioning in a more direct way, which can make learning easier in early stages. AdaGN requires the network to learn a good mapping from condition to modulation parameters, which can be harder and more sensitive to hyperparameters and training dynamics.

9 Conclusions

Across the conducted experiments with Variational Autoencoders (VAEs), including modifications incorporating perceptual loss and conditioning mechanisms, the generated outputs consistently exhibited noticeable blurriness. While perceptual loss improved semantic coherence and conditioning enhanced controllability, these adjustments were insufficient to overcome the inherent tendency of VAEs to produce over-smoothed reconstructions due to their Gaussian latent assumptions and pixel-wise reconstruction objectives. As a result, fine-grained details and high-frequency textures remained underrepresented in the generated images. Motivated by these limitations, this work extends toward Denoising Diffusion Probabilistic Models (DDPMs), which model the data distribution through an iterative denoising process rather than a single latent bottleneck. Preliminary exploration suggests that DDPMs are better suited for capturing complex image distributions and preserving high-frequency details, leading to sharper and more realistic image synthesis compared to VAE-based approaches.